Tipping-bucket rain gauges: a review of the undercatch phenomenon, and methods for its reduction and correction

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Introduction

Precipitation is a vital natural phenomenon. It is a fundamental component in the hydrological cycle, influencing both global-scale atmospheric circulation and local weather systems and ecosystems. Furthermore, its direct impact on the existence and prosperity of humankind is undeniable, as it is the primary source of freshwater. However, rainfall can be a hazard as much as a gift. In extreme situations, the occurrence of excessive rainfall can result in flash flooding, while conversely, its prolonged absence can induce drought, both posing a significant threat to life and property. The accurate measurement and monitoring of precipitation is therefore crucial, with economic, scientific and governance applications. From its role in emergency warning systems to the effective design of infrastructure and sustainable crop production and water management, rainfall data are a vital resource.

A range of instrumentation can be used to measure rainfall, and Figure 1 summarises the most prevalent. Traditionally, precipitation is measured at the Earth's surface using catching type point rain gauges, with many countries having national networks and multi-decadal records. Kidd *et al.* (2017) estimated that between 120 000 and 250 000 rain gauges were operational globally. The UK has monthly rainfall data dating back to the 1670s (Hawkins *et al.*, 2023). Initially, manual storage rain gauges were used; nowadays automatic rain gauges are widespread, with tipping bucket rain gauges (TBRGs, see Figure 2) being the most common due to their relative inexpensiveness and reliability (Strangeways, 2006; Lanza and Vuerich, 2009). In Great Britain, the Environment Agency (EA), Met Office and Scottish Environment Protection Agency (SEPA) collectively operate approximately 1300 TBRGs (EA, 2024; Met Office, 2024; SEPA, 2024).

Datasets of such magnitude and longevity enable the non-stationarity nature of precipitation trends and extremes to be studied. This provides a means of observing and assessing the impacts of climate change and variability, and for evaluating the models used to make climate change projections. Thus, despite the recent development of

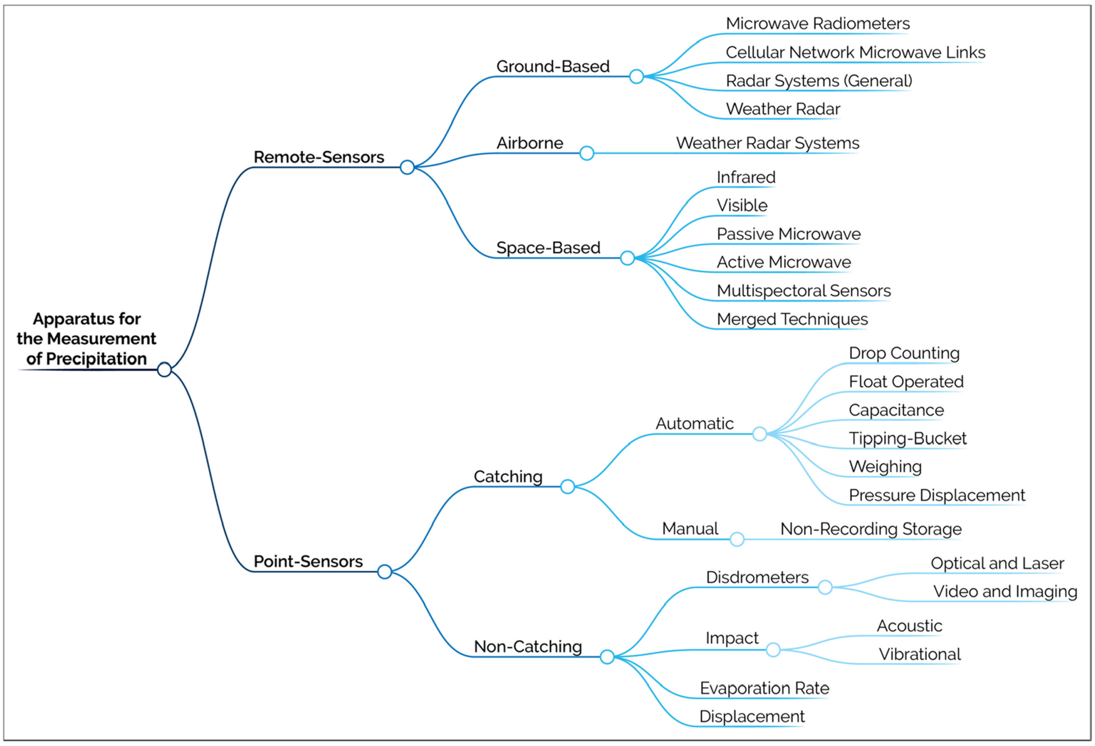

Figure 1. A schematic summarising the most commonly used apparatus for precipitation measurement. The figure framework was constructed with the aid of MarkMap.

RMetS Royal Meteorological Society

more advanced instrumentation – including weather radar and satellite-based observations – the TBRG remains essential. Additionally, remote sensing techniques still require *in situ* measurements for calibration and validation purposes, with TBRG measurements often being referred to as the '*ground truth*' (Ochoa-Rodriguez *et al*., 2019).

Unfortunately, the data collected by TBRGs is famously unreliable, being affected by a multitude of errors and typically underestimating rainfall by 5%–40% (Sypka, 2019). Since TBRGs were first invented, attempts have been made to improve the reliability of the data they collect via design and installation refinement (Duchon and Essenberg, 2001; Colli *et al*., 2018), and the development of post-collection correction methodologies (Legates and Willmott, 1990). However, despite enduring efforts, TBRG data are still compromised, an issue that is all too often overlooked (Sieck *et al*., 2007).

This paper therefore seeks to highlight the issues affecting TBRGs by summarising the key errors hindering their performance, with a focus on local errors, specifically undercatch and the different approaches implemented to reduce or correct for its effects. Previous papers have summarised the history of the TBRG and the errors that impair its performance (Strangeways, 2004; Segovia-Cardozo *et al*., 2023). However, these papers are more problem-based than solution-oriented and do not specifically focus on undercatch. In contrast, this paper describes the evolution of TBRG design in the context of undercatch reduction and explores various approaches to creating undercatch correction factors. While this paper acknowledges other methods of rainfall measurement and recognises that all catching-type instruments suffer undercatch, it focuses specifically on TBRGs due to their widespread use and the extensive

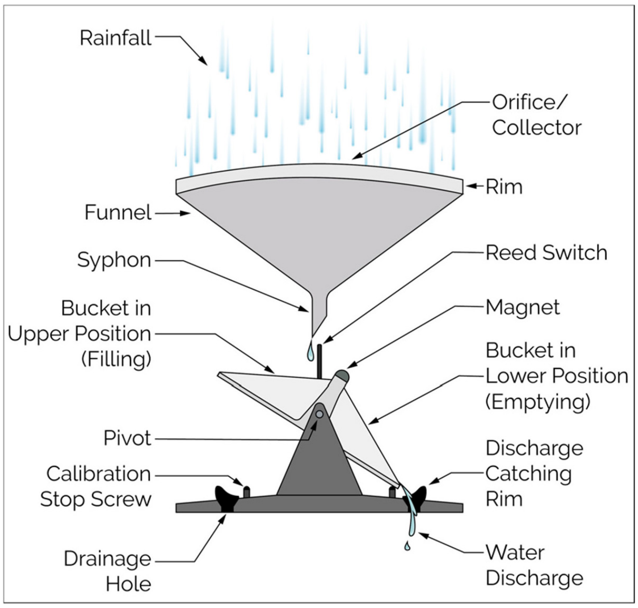

Figure 2. A diagram of the tipping-bucket mechanism. A hinge pivot in the middle enables the buckets to move in a see-saw motion, so when a bucket on one side is full, it tips down into the lower position and empties. Simultaneously, the other bucket moves into the upper position and begins filling. The buckets rock backwards and forwards for the duration of the rainfall event, the magnet between the buckets passing the reed switch each time, so each tip is automatically counted.

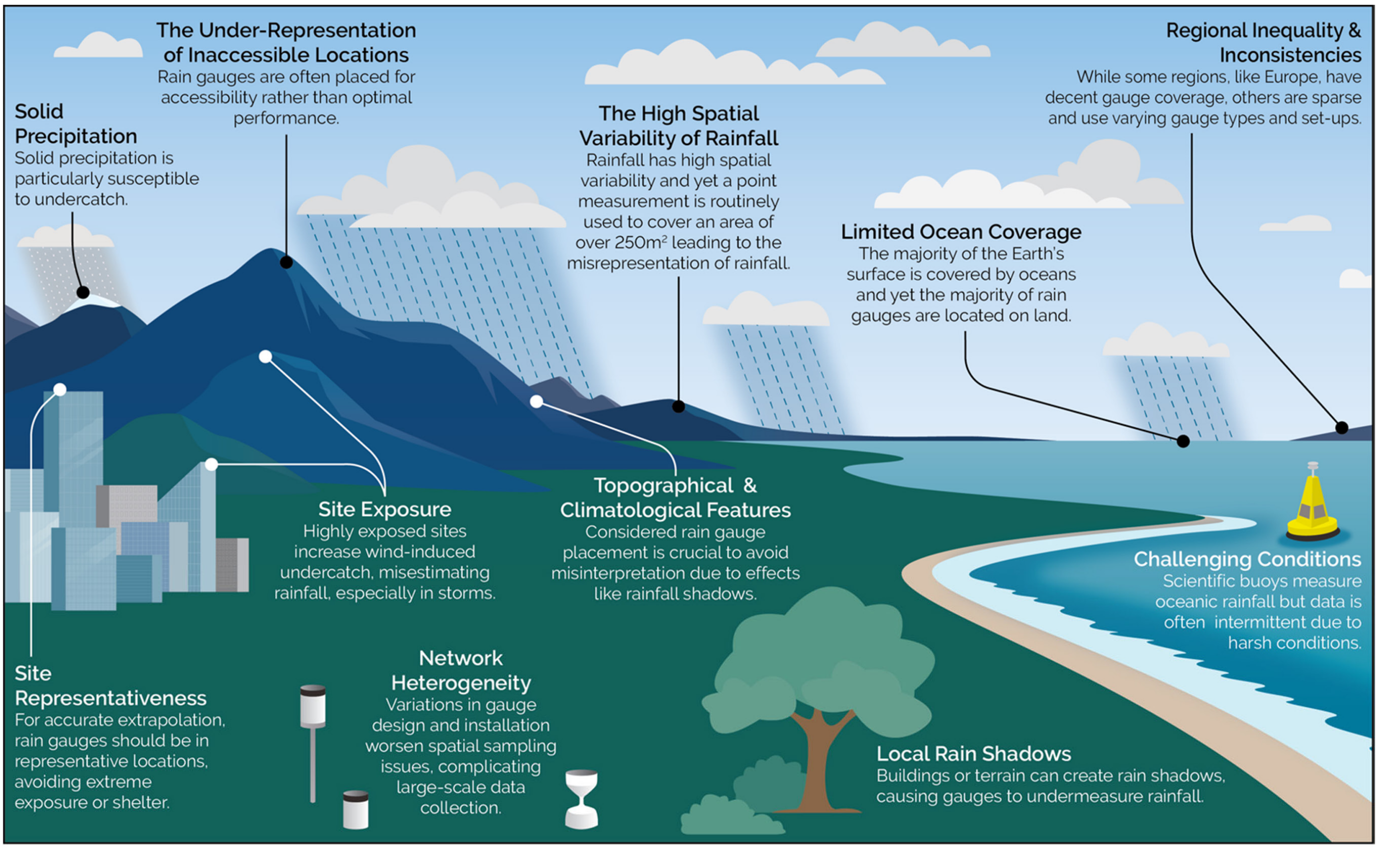

Figure 3. A pictorial summary of the primary aspects contributing to spatial sampling errors. This figure is constructed based on information from the following sources: Kidd et al. *(2017), Sucozhañay and Célleri (2018).*

body of research dedicated to them. We conclude with recommendations for future work, considering how recent technological advancements may provide a solution to this longstanding problem.

Sources of error in rain gauge measurements

TBRGs are subject to numerous error sources, impairing their performance and compromising the integrity of the applications that rely on their data. These errors are complex but, generally, can be divided into two categories: spatial sampling errors and local errors (Ciach, 2003; Villarini and Krajewski, 2008). Spatial sampling errors – due to representativeness issues when estimating precipitation over extensive areas – have been well researched and discussed (Kidd *et al*., 2017), and Figure 3 visually summarises the key factors influencing TBRG spatial representativeness. To overcome such limitations, either the number of TBRGs must be significantly increased in a strategic manner or TBRG data must be combined with another data source that better captures spatial fields, such as radar or satellite data.

The local errors impacting TBRGs are catching and counting errors (Hoffmann *et al*., 2016). Catching errors are the discrepancy between the volume of rain falling from the sky and the volume caught by the gauge, while counting errors are the discrepancy between the volume of rain caught by the gauge and the volume measured (Lanza and Stagi, 2008). The description of these errors is simple, but identifying their causations and consequences is extremely challenging. Figure 4 provides a pictorial overview of the local errors impacting TBRGs, illustrating the complexities involved.

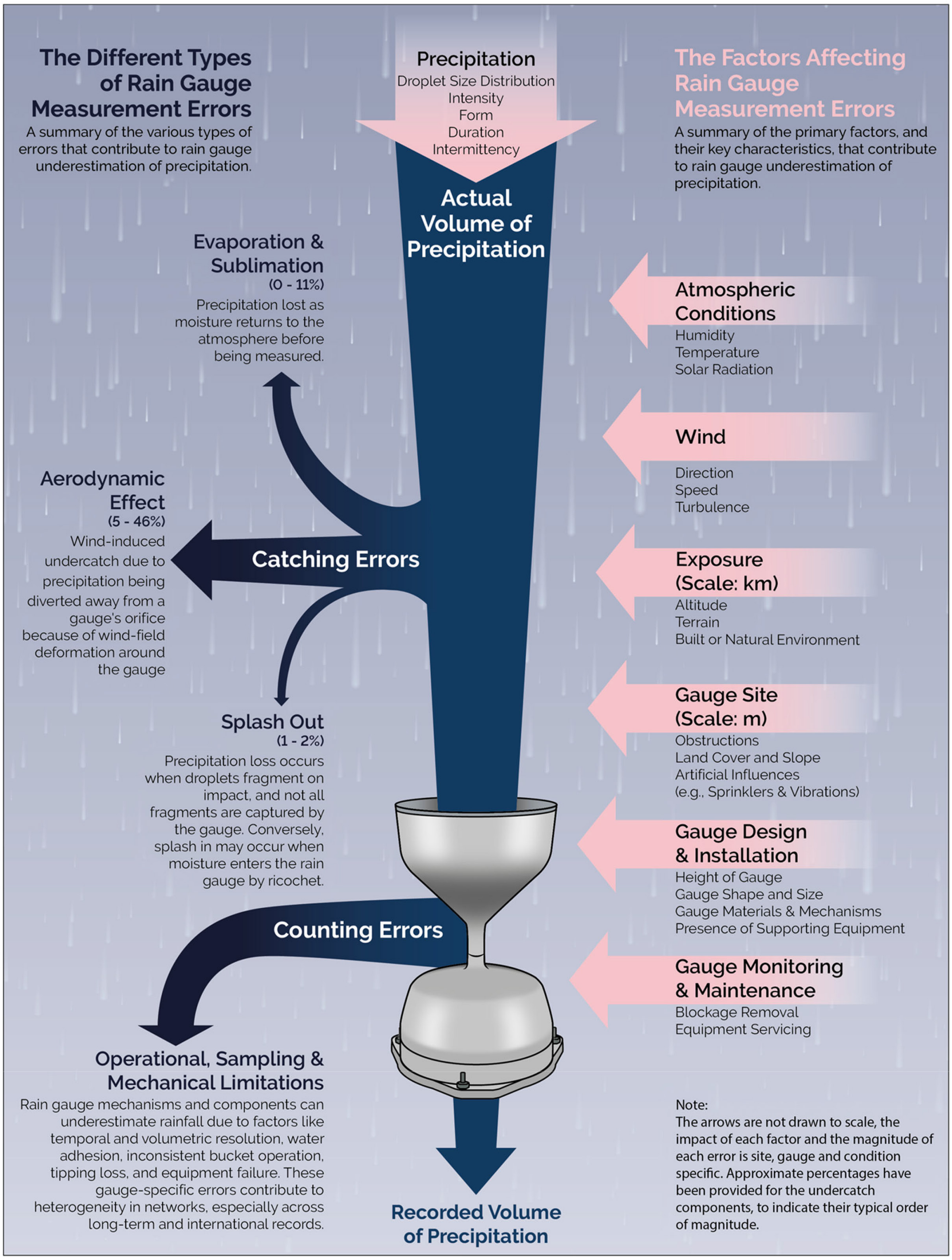

Figure 4. A conceptual model of local errors, and the factors that affect them, that result in the underestimation of precipitation by a conventional tipping bucket rain gauge. This figure is constructed based on information from the following sources: Rodda (1969), Ciach (2003), Habib et al. *(2008).*

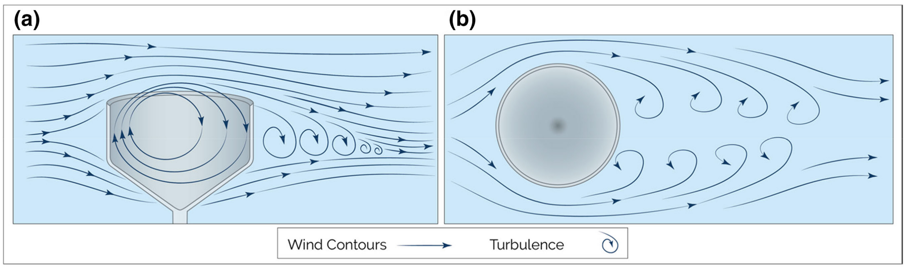

Figure 5. A diagram of bluff-body aerodynamics around an aerodynamic TBRG, (a) lateral view and (b) plan view. The drawing depicts the flow as perfect lines, but in reality, it is very unsettled. This illustration is based on information from De Smedt et al. *(2003) and Strangeways (2006).*

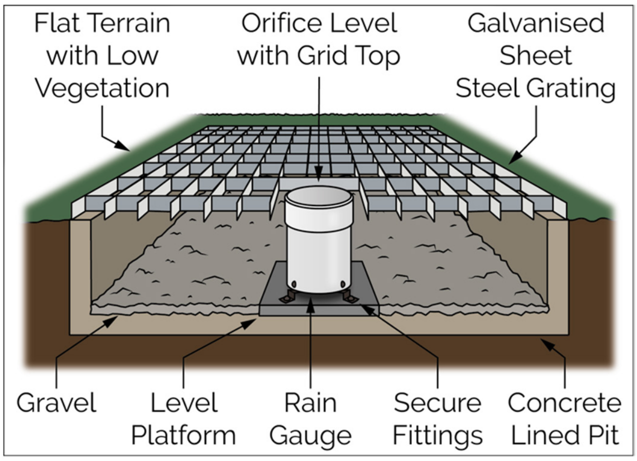

Figure 6. A diagram of a pit gauge, based on the descriptions in Rodda (1969) and Strangeways (2006). The pit enables the rain gauge orifice to be set at ground level, thereby minimising all wind effects. The grating should be strong enough to walk on without distortion and has a central open square for the correct and levelled installation of the rain gauge. To prevent in-splash, the strips of the grating are 0.3cm thick and the distance between the edge of the central square and the ground >60cm.

Both catching and counting errors can be random or systematic. Random errors encompass factors such as human error and equipment failure; hence, they are typically unpredictable, sudden and difficult to correct for. Data quality control or co-located gauges are therefore the optimum strategy for handling such errors (Ciach, 2003; Blenkinsop *et al*., 2016). Conversely, systematic errors persist over a prolonged period, typically of a consistent or gradually evolving nature. In terms of counting errors, the primary systematic errors are typically a result of a gauge's mechanical limitations or poor installation (Habib *et al*., 2001; La Barbera *et al*., 2002). Refinement of the gauge's mechanisms and adherence to installation best practices can help to mitigate such errors. Due to technological limitations, however, these errors cannot be fully eliminated. Hence, the application of a gauge-specific empirical calibration function is also recommended (Tapiador *et al*., 2012).

In general, the mechanical limitations of modern TBRGs result in a nonlinear under-estimation of rainfall, an error between 2% and 30% (Shedekar *et al*., 2016). This under-estimation decreases with increasing rain gauge resolution and increases with increasing rainfall intensity; above rainfall intensities of 30–50mmh^{-1} the reliability of TBRGs diminishes rapidly due to the saturation effect (Duchon and Biddle, 2010; Tapiador *et al*., 2012). Additionally, due to the bucket's volume dictating the minimum recording volume, information about the intermittency and intensity distribution of low rainfall rates is typically very poor (Fankhauser, 1997). Thus, the measurement principles of TBRGs significantly influence the derived statistics of rainfall extremes at both ends of the spectrum, arguably the most interesting and critical periods of rainfall (Molini *et al*., 2005). However, these errors can be minor compared to the potential 75% underestimation of rainfall during stormy conditions due to undercatch (Neff, 1977).

The undercatch phenomenon

Rainfall 'undercatch', the collective term for all catching errors, is a long-established phenomenon, with wind shown to adversely affect rainfall measurement as early as the mid-nineteenth century (Jevons, 1862; Strangeways, 2003). Subsequently, the impact of wind on gauge performance has been extensively researched (Pollock *et al*., 2018).

Wind-induced undercatch results from the airflow around the TBRG being deformed by the presence of the gauge itself, and hence is unavoidable (see Figure 5, Constantinescu *et al*., 2007). The deformation alters the trajectories of the raindrops, deviating them away from the collector (Cauteruccio *et al*., 2020). This wind-induced error is typically assessed by one of three methods: wind tunnel experiments; numerical simulation using computational fluid dynamics; and field intercomparison studies, where a pit gauge (see Figure 6) usually provides an estimation of true catch (Cauteruccio *et al*., 2024). For more modern TBRGs, wind-induced undercatch generally reduces rainfall estimation by 5%–46%, dependent on the gauge design, set-up and the ambient conditions (Duchon and Essenberg, 2001; Mekonnen *et al*., 2015; Pollock *et al*., 2018). However, this value greatly increases when precipitation is in a frozen form or when the gauge is in an exposed position. In such situations, the recorded precipitation may be less than one-third of the actual amount (Benning and Yang, 2005). Wind-induced undercatch extent is also influenced by the following: windspeed, the extent of undercatch for an unshielded rain gauge has been shown to increase by approximately 1%–2.2% for each mph increase in windspeed (Bratzev, 1963; Larson, 1971; Larson and Peck, 1974); wind turbulence, as precipitation behaves differently in uniform and fluctuating wind conditions (Cauteruccio, 2020); and the drop size distribution (DSD) of the rain, as trajectories of smaller droplets are affected more by wind than larger ones (Nešpor and Sevruk, 1999). These factors all have high spatiotemporal volatility, and their influence is gauge specific. Thus, wind-induced undercatch is challenging to quantify and to correct for.

Wind-induced losses are generally considered to be the most significant error contributing to undercatch and hence are the most thoroughly investigated (Rodda, 1969; Yang *et al*., 1999), but wind is not the only source of undercatch. Evaporation, sublimation and out-splash also influence gauge undercatch extent. sublimation and evaporation-induced undercatch occur when precipitation collected in the funnel vaporises without being measured. These processes are highly dependent on the local climatology; in cooler regions, evaporation errors are often considered negligible (Yang and Ohata, 2001); however, in warmer regions they can reduce measured rainfall by 4%–11% (Zhang *et al*., 2004). The opposite is true for sublimation: this error is irrelevant in warmer regions, but where snowfall is present, it may reduce daily rainfall volumes by 0.5–0.75mm (Fassnacht, 2004). The rates of sublimation and evaporation

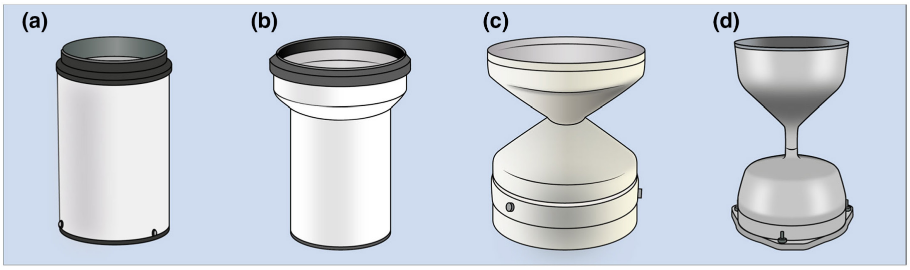

Figure 7. A visual summary of the most common TBRG shapes. (a) A conventional cylindrical shape. (b) A cylindrical shape with a funnel top. (c) An aerodynamic (plastic) shape, developed at the Institute of Hydrology. (d) An aerodynamic (aluminium) shape also referred to as calix-shape or champagne glass shape (Strangeways, 2004).

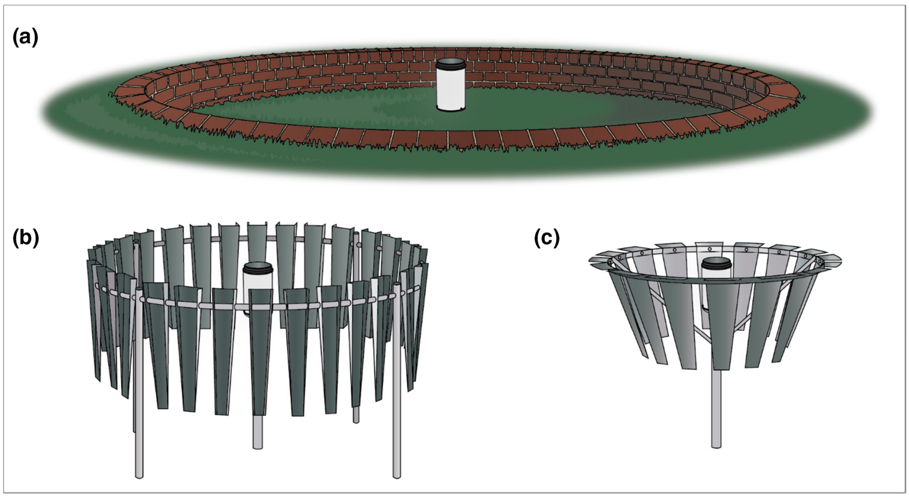

Figure 8. Examples of the different supporting features that can be used to reduce wind-induced undercatch. (a) A turf wall, the height of the wall is the same as the height of the rain gauge. (b) Alter windshields with hanging vertically oriented slates, double shield versions are also available. (c) OTT screen style.

are influenced by a variety of external factors, such as air humidity, temperature and velocity, and are themselves difficult to measure. Consequently, evaporation and sublimation-induced undercatch are also hard to correct for.

The final, and typically least significant, undercatch error is out-splash. Splashing losses refer to the water lost when a droplet fragments on impact and is not entirely captured by the TBRG's collector. This error can result in approximately 1%–2% undercatch (Cauteruccio *et al.*, 2021). However, most contemporary gauges have a deep funnel, generally resulting in this error being negligible (Legates, 1992).

Undercatch is the most challenging error impacting the reliability of TBRG measurements due to its multifaceted and dynamic nature (Figure 4, Sieck *et al.*, 2007). Furthermore, undercatch is very location-specific in both extent and causation (Ye *et al.*, 2004; Buisán *et al.*, 2017; Herrnegger *et al.*, 2018). This complexity is further compounded in temperate climates, as undercatch often exhibits seasonality (Zhao *et al.*, 2024). To what extent a TBRG's performance is influenced by these external forces is strongly dependent on the site's level of exposure and the gauge's design and installation (Colli *et al.*, 2018; Pollock *et al.*, 2018).

Undercatch reduction

The two fundamental strategies for reducing undercatch errors are improved gauge design and considered gauge installation (Segovia-Cardozo *et al.*, 2023). Initially, TBRGs were cumbersome pieces of apparatus; however, advancements in manufacturing processes and materials have led to more robust and refined gauge design (Cornish and Green, 1982). Typically, early TBRGs were a cylindrical shape, while newer gauges have an aerodynamic profile (Figure 7). This shape reduces airflow deformation and hence undercatch extent (Colli *et al.*, 2018; Cauteruccio *et al.*, 2024). The calix-shape rain gauge is currently considered the most optimal design for reducing undercatch (Folland, 1988; Pollock *et al.*, 2018).

Additionally, more subtle advancements have been made in gauge design. Orifice rim thickness and profile have been shown to influence catch efficiency (Cauteruccio *et al.*, 2021). Other design elements that have been identified to influence undercatch extent are the following: the deepness of the collector's vertical wall; the slope of the funnel; and the funnel's coating (WMO, 1996; Mekis and Vincent, 2019; Padrón *et al.*, 2020). Notably, not all advancements are unequivocally beneficial. For example, in arid regions, gauges with larger funnel sizes are often used because they are more effective at detecting light rainfall events of <1mm. However, this design alteration can introduce a bias by underestimating the depth of high-intensity storms exceeding 10mm (Al-Wagdany, 2015). It is therefore important to consider both the intended and unintended impacts of any gauge design modifications.

Improvements in installation practices have evolved alongside improved gauge design. It has been demonstrated that instrumentation height influences undercatch extent (Pollock *et al.*, 2018), with higher TBRGs experiencing stronger effects from wind (Mekonnen *et al.*, 2015). Consequently, TBRG's in the United Kingdom are now installed at ground level, thus reducing wind exposure and creating measurement consistency across the network (Muchan and Dixon, 2019). However, this practice is not necessarily replicated in other countries, and does increase the risk of debris being blown into the orifice and splash-in (Sieck *et al.*, 2007). In some instances, supporting features such as wind shields or turf walls have been used to reduce undercatch: Figure 8 features examples of these structures. However, such options are not always practical (Duchon and Essenberg, 2001). Additionally, nowadays greater consideration is typically given to a gauge's siting in terms of its surrounding terrain and its proximity to obstructions (Suri and Azad, 2024). The recording of detailed metadata on gauge siting is crucial, both with regards to gauge type and site description, to capture these discrepancies and provide context to measurements. Intercomparison studies have highlighted the large variability among individual TBRGs, both in field settings and under controlled laboratory conditions (Lanza and Stagi, 2008; Sevruk *et al.*, 2009). Awareness of these inconsistencies and their implications is essential, especially when using transnational or multidecadal datasets, where network heterogeneity is almost inevitable.

Undercatch correction

Undercatch cannot be eliminated, and network heterogeneity is inevitable. Therefore, correction factors (CFs) are required to estimate the actual volume of rainfall at the gauge. This concept is not a novel one: since the 1980s, various attempts have been made to develop an effective CF for TBRGs. Table 1 provides a

Table 1

A summary of some of the key methodologies that have been developed for correcting rain gauge catching errors.

Name and method (country)	*Spatial resolution*	*Temporal resolution*	*Required data*	*Advantages*	*Disadvantages*	*References*
Statistical analysis model Analysing the ratio of daily amounts of precipitation measured at ground level and at standard height (1.5m) (Denmark)	Gauge	Daily	• Air temperature • Exposure level • Wind speed (10m above ground level) • Precipitation intensity	• Allows for three different levels of exposure, well-sheltered, moderately sheltered or unsheltered • Considers both solid and liquid precipitation, though only in a simplistic binary manner • Considers aerodynamic effects and wetting loss • Relatively easy to apply with clear tabulated values	• Only considered the Hellmann rain gauge • Based on the assumption that exposure errors at ground level are negligible • Wind speed measurements are not always available at each TBRG site	Allerup and Madsen (1980)
Legates correction factor (CF-L) Gauge measurement errors were estimated, these irregularly distributed records were then interpolated to a 0.5 of latitude by 0.5 of longitude grid using a spherically based interpolation algorithm (USA, Global Data)	≈238m²	Monthly	• Location • Date	• Simplicity, a single correction factor that can be applied to any rain gauge based on the gauge's location and month • Universality, the developed correction factors provide global coverage and are not gauge specific • Is design to correct for errors due to the effects of wind, wetting losses and gauge evaporation	• The CF-L may be outdated, as it relies on data from 1920 to 1980s • Arid, mountainous and polar regions were underrepresented in original data sources • Spatially gridded CF-Ls do not capture the high spatial variability of topography, and the built or natural environment • Meteorological parameters have only been roughly estimated which may lead to an over-correction effect • This approach does not allow the impact of gauge design to be effectively captured	Legates and Willmott (1990), Ehsani and Behrangi (2022)
Dynamic correction model A statistical model using regression analysis techniques with log-values (Norway, Data from Finland)	Gauge	Daily	• Precipitation intensity • Wind speed measured at gauge level • Precipitation amount • Gauge type • Air temperature at 2m	• Droplet size distribution (DSD) is incorporated, although it is assumed based on rainfall intensity • The crystal structure for solid precipitation is incorporated as a function of air temperature • Site exposure is part of the correction formula	• Designed for four particular gauges with a specific windshield set-up and is only applicable in these circumstances due to the use of tabulated parameters • Evaporation and wetting loss were developed in a Nordic context and are simply provided as set monthly values	Førland *et al.* (1996), Michelson (2004)
Numerically based correction procedure The wind-induced error is characterised by a non-linear complex behaviour dependent on wind speed, rainfall rate and DSD (USA)	Gauge	Resolution of recorded data (1min to 1 month)	• Drop size distribution (DSD) • Wind speed • Precipitation intensity	• DSD is factored by the inclusion of rain type as a parameter (orographic, thunderstorm or showers) • The correction can be applied at different timescales, reducing the likelihood of over correction	• Only wind-induced errors were considered • Designed for a specific class of rain gauge types (Mk2) and so the formula can only be applied to other gauges of a similar geometry and operational principles	Habib *et al.* (1999)

(Continues)

Table 1 (Continued)

Name and method (country)	*Spatial resolution*	*Temporal resolution*	*Required data*	*Advantages*	*Disadvantages*	*References*
Explicit mathematical statistical model Correction factors derived from known values of four independent variables (Denmark)	Gauge	Daily	• Precipitation intensity (remote) • Snow fraction (remote) • Wind speed (remote) • Temperature (remote)	• On-site observations are not required for wind speed, temperature and snow fraction, instead remote measurements can be utilised. Wind speed can be extrapolated from sites not farther than 50km, while rain intensity and temperature can be extrapolated across greater distances • May require a long period of measurements to establish a reliable correction model	• Its limited versatility, this correction methodology only works for the Hellman rain gauge and was developed in a Danish context • The theory was only tested on 12 rain gauges and did not consider different levels of exposure • The remote observations still need to be within a certain distance of the gauge, maximum distances of 50–100km depending on the variable	Allerup *et al.* (2000)
Fuch's dynamic correction method (CF-F) Based on the dynamic correction model described in Førland *et al.* (1996) (Germany and Austria)	Gauge	Daily	• Air temperature • Dew point temperature • Relative humidity • Wind speed at gauge rim • Precipitation intensity	• Allows for inconsistent forms of precipitation across each time increment, by incorporating the Global Precipitation Climatology Centre (GPCC) phase scheme and using this to weight the correction factors for a certain station and day	• This method is based on Førland *et al.* (1996), hence it has inherited most of the underlying method's issues • In tropical regions especially the calculated precipitation intensities could be too low, which may lead to over-correction effects • Requires a lot of input data, and wind speed at gauge rim is a relatively unusual measurement to have available	Fuchs *et al.* (2001)
Catch ratio method Catch ratio as a function of daily mean wind speed, using separate regression equations for snow and rain (China)	Gauge	Daily	• Air temperature • Precipitation intensity • Wind speed (10m above ground level)	• Allows for mixed precipitation estimated by a linear relationship of snow and rain percentages determined by daily temperatures • This methodology also considers trace precipitation, wetting losses and wind-induced errors	• This method does not incorporate gauge design or exposure • The bias correction is based on certain conservative assumptions such as mean amounts of trace and wetting losses • Evaporation losses are not considered	Ye *et al.* (2004)
Calculated relative differ-ences A simplified equation developed based on the relationship between logarithmic wind profile and its effect on the catch ratio (Czech Republic)	Gauge	Daily	• Wind speed • Precipitation intensity	• Detailed consideration of wind-induced losses, wetting losses, evaporation losses and trace amount of precipitation during the development of the correction method • Wind speed is the only additional measurement required to implement the correction procedure	• The methodology was developed based on a single study site, and for specific gauge types and set-ups • Only applicable to liquid precipitation • Different exposer levels due to varying terrain are not considered	Mekonnen *et al.* (2015)
Overall collection effi-ciency method Wind-induced under-catch quantified using Computational Fluid Dynamics with embedded liquid and solid particle tracking (Italy)	Gauge	Resolution of recorded data	• Precipitation intensity • Droplet size distribution (DSD) • Precipitation form • Wind speed	• A highly detailed investigation exploring the effect of gauge design on wind-induced undercatch extent	• A relatively complex correction equation to be applied • Developed in highly controlled laboratory conditions and not designed explicitly for real-world application • Only wind-induced undercatch was assessed	Cauteruccio *et al.* (2024)

chronological summary of a selection of these attempts, with their key strengths and weaknesses highlighted. CFs were chosen not only because they are some of the most widely adopted, but also to emphasise the challenges involved and the diverse approaches adopted.

The first decision when creating a CF is what independent, accurate and representative reference measurement should be the '*ground truth*' (Michelson, 2004). Early CFs, such as Allerup and Madsen's (1980), used ground-level rain gauges; however, more recent studies typically use pit gauges (Duchon and Essenberg, 2001), shielded gauges (Kochendorfer *et al*., 2017), laboratory experiments (Nešpor and Sevruk, 1999) or numerical simulations (Cauteruccio *et al*., 2024). Each of these approaches has their own limitations. With field studies, it is difficult to isolate individual effects as there are so many uncontrolled variables. Furthermore, the true precipitation value is not known as both pit and shielded gauges are still susceptible to errors (Michelson, 2004; Devine and Mekis, 2008). Conversely, theoretical models and laboratory experiments allow all variables to be controlled for, and hence a detailed investigation of individual factors can be conducted (Cauteruccio, 2020). However, consistent artificially-created variables do not replicate real-world conditions, and therefore, the findings can be difficult to apply to operative rain gauges. Additionally, such methods often involve controlling parameters, like DSD and windspeed at the gauge rim (Cauteruccio *et al*., 2024), that are not recorded at most TBRG sites. Therefore, while such factors have been demonstrated to influence undercatch extent, their direct inclusion in a CF is highly limiting (Cauteruccio *et al*., 2023).

A potential substitute for missing on-site observations is to infer these variables from other available data sources. At a basic level, this might be using temperature as a proxy for precipitation form (Allerup and Madsen, 1980) or precipitation intensity as an indicator of DSD (Cauteruccio *et al*., 2023). Alternatively, gridded or remote data can be converted into gauge-specific variables via a reduction equation or a statistical model (Allerup *et al*., 2000). Implementing such measures enables the application of more detailed CFs, even at data-poor sites. However, it is important to consider if such data can capture what is occurring at the gauge level or if it will lead to the misrepresentation of variables.

CFs depend not only on the characteristics of individual precipitation events, but also on other environmental factors (Yang *et al*., 1998). Hence, detailed metadata on instrumentation, set-up and siting should ideally be available as the quality of the CF is contingent on the provision of this information (Fuchs *et al*., 2001). Unfortunately, missing, incomplete or inconsistent metadata is a common issue, thereby preventing observations from being put into proper contexts and potentially introducing significant biases (Aguilar *et al*., 2003).

As stated in Table 1, many CFs have been created for a specific gauge with a specific set-up and in a specific environment, thereby limiting their transferability. As gauge construction and installation have a significant impact on undercatch extent (Pollock *et al*., 2018), this is therefore the more conservative approach. Alternatively, CFs can be designed to be versatile, with even global coverage of a gridded nature, such as Legates and Willmott (1990). However, using a universal CF or a CF with a coarse temporal scale, while being easy to apply, can lead to the over-correction of precipitation by up to 200% (Ehsani and Behrangi, 2022; Cauteruccio *et al*., 2024), and hence such strategies should be used with caution. Adopting a categorical approach to certain variables may be the optimal compromise: for example, Rubel and Hantel (1999) developed a different CF for liquid, mixed and solid precipitation. Furthermore, this approach could also be applied to allow for varying site characteristics: for example, Allerup and Madsen (1980) incorporated exposure into their CF by allowing a site to have a one to three grading for this characteristic. However, this oversimplification may result in the misrepresentation of the site conditions and harm the integrity of the CF, so such approaches should only be used after thorough investigation and testing.

Rain gauges have become more precise in measuring over time, and therefore less data adjustment is now needed (Mekis and Vincent, 2019; Joelsson *et al*., 2024). Nonetheless, if heterogeneous TBRG datasets are to be used in research, developing an effective CF that can be easily and broadly implemented is essential to ensure continuity across all observations (Scaff *et al*., 2015). Currently, most available datasets for global hydrological simulation do not account for rain gauge catch deficiencies (Adam and Lettenmaier, 2003), and existing CFs result in significantly different rainfall estimations (Ehsani and Behrangi, 2022).

Future directions for undercatch correction

With the growing prevalence of artificial intelligence methods and 'big data', new possibilities now exist for the correction of TBRG data. For example, spaceborne sensors provide spatially expansive data that can be used to validate and refine gauge undercatch correction methods (Behrangi *et al*., 2018). Additionally, machine learning models have facilitated the merging of heterogeneous data sources to remove biases from TBRG datasets (Guarascio *et al*., 2020), and could be used for leveraging large and complex datasets to identify patterns that previously more traditional techniques have been incapable of recognising.

Technological advancements have also led to the evolution of the TBRG itself, with new research indicating that small gauge-mounted sensors may facilitate the real-time quantification of undercatch (Dutton and Balsamo, 2024). Additionally, there is a need to develop an alternative method for establishing the '*ground truth*' of TBRG measurements, as even the most reliable instruments – such as pit gauges – underestimate actual rainfall (Devine and Mekis, 2008). Rainfall simulators have been used in soil erosion and runoff studies (Abudi *et al*., 2012), and potentially could be adapted towards researching undercatch in field experiments where the true rainfall is known and controlled for. Consequently, innovative approaches and new technologies may provide a means to solving the longstanding problem of rain gauge undercatch.

Conclusion and recommendations

The TBRG is one of the most widely used tools for measuring rainfall, aiding in the observation and monitoring of the Earth's current climate and the understanding and assessment of its changes. However, this fundamental device is susceptible to a myriad of errors which have been found to influence trends and lead to the mismanagement of water resources and disasters (Archer *et al*., 2007; Li *et al*., 2023). Thus, there is a crucial need for the application of an undercatch CF to make rain gauge data more reliable (Sevruk *et al*., 2009). Various CFs have been developed, from the Legates monthly correction factor that can be applied to any TBRG (Ehsani and Behrangi, 2022) to instrumentation-specific ones that can only be used on certain gauges in particular settings (Rubel and Hantel, 1999). While existing CFs have been demonstrated to be successful in their improvement of precipitation data (Feng *et al*., 2024), further research is still required in this area, with no singular CF being widely applied or accepted, and existing CFs resulting in significantly different rainfall estimations (Ehsani and Behrangi, 2022). Without greater understanding and awareness of undercatch, TBRGs and the integrity of the data they collect will continue to be compromised.

We recommend:

- That future studies incorporating rain gauge data should acknowledge the impacts of undercatch, either by recognising the uncertainty it introduces or by applying an appropriate CF;

- Further research into rain gauge accuracy, using a rainfall simulator to ensure the *'ground truth'* is known;
- The development of a CF, specifically for a UK context, that is versatile enough to be widely adopted but not to the extent that it leads to the misrepresentation of conditions and undercatch overestimation.

Author contributions

Ruth E. Dunn: Visualization; writing – original draft; writing – review and editing; investigation; conceptualization. **Hayley J. Fowler:** Conceptualization; supervision; writing – review and editing. **Amy C. Green:** Conceptualization; supervision; writing – review and editing. **Elizabeth Lewis:** Conceptualization; supervision; writing – review and editing.

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Data availability statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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